

Random Weather Generator

Process Design Document (PDD)

Raya Information Technology
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The logo for Raya Information Technology, featuring the letters 'RAYA' in a stylized, bold, blue font. The 'A's are unique, with the first having a small triangle above it and the second having a small triangle to its right. The logo is set against a white square background.

Process Design Document History

Date	Version	Role	Name	Organization	Function	Comment
08.09.2025	1.0	Author	Mohamed Mesbah	N/A	SME	Creation v 1.0
09.09.2025	1.1	Reviewer	Yasser Salah	Raya	BA	Approved v 1.0

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1. Objective

The objective of this process is to generate synthetic weather data by creating randomized yet realistic values for temperature, humidity, wind speed, precipitation, and weather conditions. This generated data can be used for testing, simulations, and demonstrations without depending on real-world weather APIs.

By introducing Robotic Process Automation (RPA), the process aims to:

- Eliminate the need for manual creation of test weather data, which is repetitive and time-consuming.
- Increase the speed and scalability of generating large volumes of synthetic weather records.
- Free up employees to focus on analysis, testing design, and higher-value tasks rather than routine data preparation.

1.1 Business Objectives and Benefits

Objectives	Expected Benefits
Automate the creation of artificial weather records such as temperature, humidity, wind, and precipitation.	Saves employees from repetitive and time-consuming manual data preparation.
Standardize how weather data is generated, stored, and distributed through automation.	Improves consistency and accuracy across all test datasets.
Enable large-scale generation of weather scenarios for different use cases.	Increases scalability and processing speed for handling high data volumes.

2. Current Process

2.1 Process Overview

The process begins when the user accesses the Random Weather Generator website. The user selects the required parameters (Climate, Season, or Supernatural Weather) and submits the request. The website generates random weather information including Description, temperature, wind, precipitation, and overall conditions. The results are displayed on the screen and can be copied for use in testing, simulations, or demonstrations.

With RPA, the bot automates this interaction by opening the website, entering the parameters, generating the weather data, extracting the results, and saving them in the required format. This removes manual effort and ensures fast, repeatable, and accurate data collection.

2.2 Standard of Metrics

General information about the process selected for RPA, prior to automation:

Process name	Random Weather Generator
Function	generate randomized weather data
Department	IT / Quality Assurance (Test Data Management)
Process short description	The process retrieves randomized weather data from the Random Weather Generator website by selecting parameters (climate, season, ...). The results are extracted and stored for use in testing, simulations, or demonstrations.
Process schedule	On-demand or scheduled as needed (e.g., daily, weekly, or before specific testing/simulation activities).
#of items process/month	Approximately 200–500 weather records, depending on testing and simulation requirements.
Average cycle time	2–3 minutes per run.
Average handling time	1–2 minutes of manual effort per run before automation (reduced to near zero with RPA).
Peak period(s)	N/A
# of FTE supporting this activity	1 FTE (part-time effort, primarily for data preparation and validation before automation).
Input Data	Parameters selected on the Random Weather Generator website (climate, season, supernatural weather).
Output Data	Weather Properties (Description, Temperature, ...)
Return on investment	High ROI due to elimination of manual data preparation, faster generation of large datasets, reduced errors, and improved efficiency in testing and simulation processes.

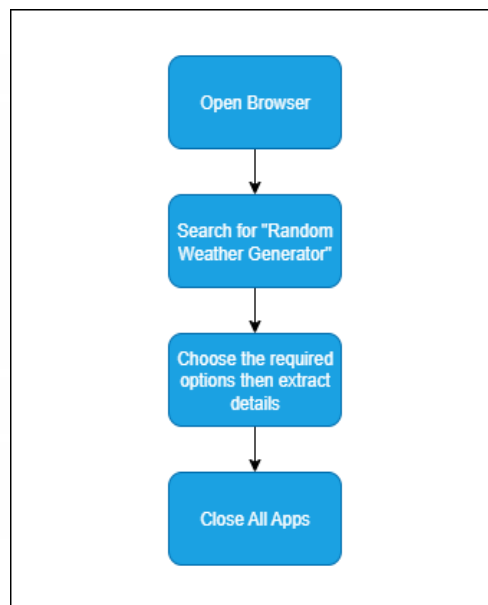
2.3 Volume of data

Activity	Time taken per month (hours)	Volume of data to be processed
Accessing Random Weather Generator website	3–4 hrs	~200–500 records
Selecting parameters	2 hrs	~200–500 records
Generating weather data	4–5 hrs	~200–500 records
Extracting results	6–7 hrs	~200–500 records
Formatting and storing data	8–10 hrs	~200–500 records

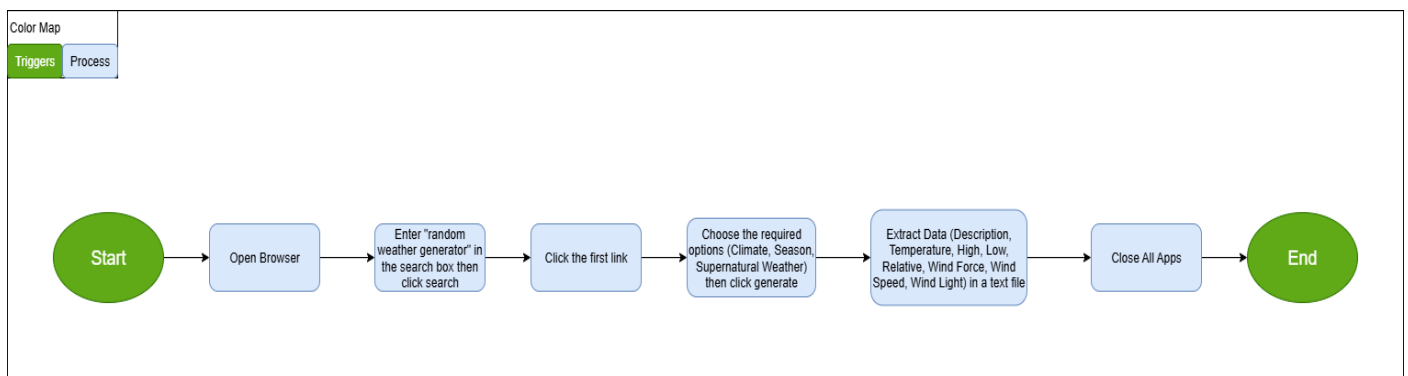
3. Business Rules

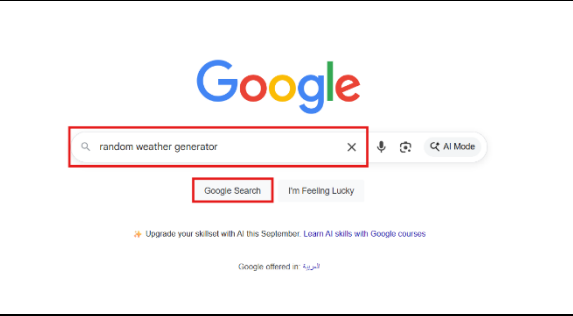
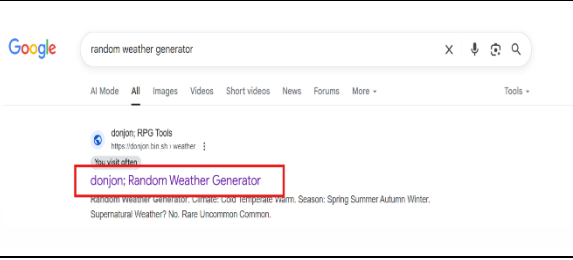
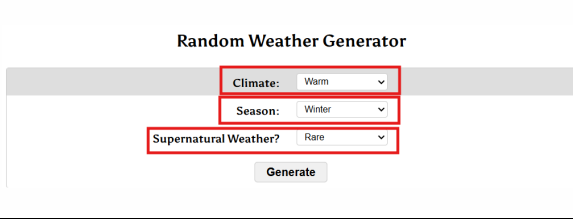
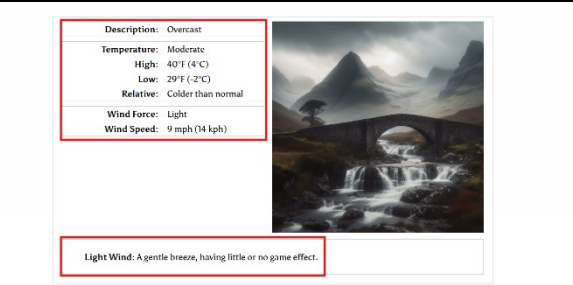
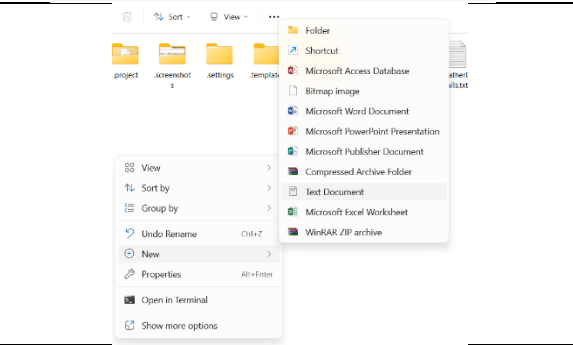
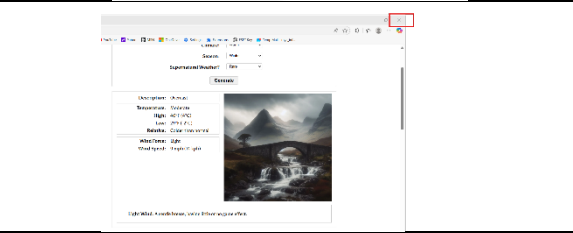
BR#	Requirement	Additional Information
#BR001	User (or bot) must access the Random Weather Generator website.	This is the starting point of the process.
#BR002	Parameters (season, climate, and Supernatural Weather) must be selected before generation.	Ensures the weather data reflects the intended scenario.
#BR003	The “Generate Weather” button must be clicked to produce results.	Triggers the creation of randomized weather data.
#BR004	Weather output must be extracted from the website results.	Data is copied/downloaded for further use.

4. High Level Process Diagram



5. Detailed Process Steps – AS IS



Steps	Description	Screenshots	Exception Handling
1	Open Browser		#SE002
2	Enter "random weather generator" in the search box then click search		
3	Click the first link		#SE001
4	Choose the required options (Climate, Season, Supernatural Weather) then click generate		
5	Extract Data (Description, Temperature, High, Low, Relative, Wind Force, Wind Speed, Wind Light)		#BE001
6	Open Text File (Notepad) To Extract data into		
7	Close All Apps		

6. In Scope for RPA

The activities and exception in of scope for RPA, in this automation workflow /sequence are listed here:

- Full Scope of the process for RPA - to be 100% automated

7. Out of Scope

- There are no activities out of scope for RPA ...

8. Non-process requirements

8.1 Systems

System	Usage	Frequency
Edge Browser	To Open the required website	Daily (multiple times per day)
Notepad	To store the extracted data	Daily / As required per work item

8.2 Reporting

Report name	Source	Purpose
Weather Data Report	Random Weather Generator website output	Provides generated weather details for testing or simulations.
Process Execution Log	RPA tool (e.g., UiPath)	Tracks process runs, parameters used, and execution status for audit and monitoring.
Error/Exception Report	RPA tool and system logs	Highlights failures such as website unavailability or data extraction issues, enabling quick resolution.

8.3 Archiving

Generated weather data and process logs must be stored securely in designated folders or databases for future reference and audit purposes, with retention based on organizational policies.

8.4 Security & data retention

Access to the Random Weather Generator process and its outputs must be restricted to authorized users. All generated datasets and logs should be retained securely for a defined period (e.g., 12 months) in compliance with organizational policies and then archived or deleted as required.

9. Exception Handling

9.1 Business Exception

#Business Exception no.	Business Exception	Description	Expected Action
#BE001	Incorrect data format	Extracted results are not aligned with the expected format (e.g., missing columns, invalid values).	Raise exception for review.

9.2 System/Application Exception

# System Exception no.	Exception	Description	Expected Action
#SE001	System Downtime / Timeout Error	The website is offline or under maintenance.	Retry for 3 times. If still unavailable → raise System Exception.
#SE002	Network connectivity issues	Loss of internet prevents access to the website.	Retry connection; alert IT if persistent; pause automation until connection is restored.

9. Escalations

For immediate contacts for the process automation project, kindly contact the respective contact on the table below:

Email Address	Roles	Responsibilities
rpa-team@company.com	RPA Developer	Handle bot failures, update automation logic, and maintain process continuity.
qa-lead@company.com	QA / Test Lead	Validate generated data, review exceptions, and ensure process meets testing requirements.
support@company.com	IT Support	Resolve website access, network, or system-related issues.